

Complete Requirements Table

Pharmacy Expiry & Re-order Dispatch Engine | Problem Statement #15

Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Pharmacy Clerk to record each received medicine batch with medicine identifier, batch number, quantity, received date, expiry date, and supplier.	High	Pass: A valid batch is saved and is available in inventory with all six fields. Fail: The system accepts a missing batch number, quantity, or expiry date.	Batch-level data is essential for expiry tracking, traceability, and FEFO selection.
FR-002	Functional	The system shall identify and display batches that are expired or will expire within a configurable alert period.	High	Pass: On the daily check, all expired batches and batches within the configured period are listed with status and expiry date. Fail: A qualifying batch is absent from the alert list.	Early visibility prevents unsafe dispensing and reduces wastage.
FR-003	Functional	The system shall generate a FEFO dispensing list that orders only unexpired batches of a selected medicine by nearest expiry date first.	High	Pass: The nearest-expiring valid batch is first; expired batches are excluded. Fail: A newer-expiry batch precedes an older valid batch.	FEFO minimizes expiry-related loss while ensuring patients receive safe stock.
FR-004	Functional	The system shall compare the available stock of each medicine against its configured safety threshold after every stock receipt and dispensing transaction.	High	Pass: A threshold breach is detected immediately after a transaction lowers available stock below the threshold. Fail: A below-threshold medicine is not flagged.	Continuous threshold checking prevents stock-outs of essential medicines.
FR-005	Functional	The system shall automatically generate a purchase order when medicine stock falls below the configured safety threshold and dispatch it to the assigned Inventory Supplier after Pharmacy Clerk approval.	High	Pass: An approved order contains medicine, calculated re-order quantity, supplier, and order reference, and the supplier receives it. Fail: An order is dispatched without approval or missing required details.	Automating re-order preparation shortens replenishment time while retaining clerk control.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance	Under a simulated peak load of 100 concurrent clerk requests, the system shall return a FEFO dispensing list or threshold-check result within 2 seconds for 95% of requests.	High	Pass: Load-test results show the 95th-percentile response time is at most 2 seconds. Fail: The 95th percentile exceeds 2 seconds.	Pharmacy operations require prompt recommendations at checkout and during busy periods.
NFR-002	Security	The system shall require authenticated Pharmacy Clerk access for inventory changes and purchase-order approval, and shall keep an audit record of the user, timestamp, and action for at least 12 months.	High	Pass: Unauthenticated changes/approvals are denied; each authorized action has a complete audit record retained for 12 months. Fail: An unauthenticated action succeeds or an audit record is incomplete.	Inventory and procurement records are sensitive operational data and need accountability.